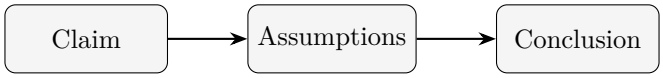
DISCRETE STRUCTURES

CMSC250

Chapter 8: The Language of Proofs

By Naren Ayinala

Definitions, hypotheses, conclusions, and counterexamples



Contents

1 What is a mathematical proof? 3

2 The structure of a conditional claim 3

3 Definitions are tools, not decorations 3

4 Universal and existential statements 3

5 Closure properties and the right level of detail 4

6 Proof hygiene and common errors 4

7 Practice problems 4

8 Solutions to practice 4

1 What is a mathematical proof?

A proof is a finite chain of statements showing that a claim follows from agreed definitions, earlier theorems, and valid rules of reasoning. A convincing numerical pattern is not a proof of a universal claim. For example, checking that $n^2 + n$ is even for $n = 1, 2, 3$ suggests a pattern; it does not establish the result for every integer.

Key idea. A proof explains why a statement must hold for every object covered by its quantifiers. Every new claim should have a reason.

2 The structure of a conditional claim

To prove $P \to Q$, assume P and derive Q. Do not assume Q at the outset. To disprove $P \to Q$, find one case where P is true and Q is false. In a proof, write explicitly what is assumed and what remains to be shown.

Worked example: an implication about multiples

Claim: For every integer n, if $6 \mid n$, then $3 \mid n$. Let $n \in \mathbb{Z}$ be arbitrary and assume $6 \mid n$. By the definition of divisibility, $n = 6k$ for some $k \in \mathbb{Z}$. Thus $n = 3(2k)$. Since $2k \in \mathbb{Z}$ by closure of integers under multiplication, the definition of divisibility gives $3 \mid n$. Since n was arbitrary, the claim holds for every integer n.

3 Definitions are tools, not decorations

The statement "n is even" means $n = 2k$ for some $k \in \mathbb{Z}$; "n is odd" means $n = 2k + 1$ for some $k \in \mathbb{Z}$. A rational number has the form a/b with $a, b \in \mathbb{Z}$ and $b \neq 0$. The nonzero condition is part of the definition and must remain true after algebraic transformations.

Worked example: sum of two even integers

Let a, b be arbitrary even integers. By definition, $a = 2r$ and $b = 2s$ for integers r, s. Then $a+b = 2(r+s)$. By closure of $\mathbb{Z}$ under addition, $r + s \in \mathbb{Z}$. Hence $a + b$ is even by definition. This proof does not merely show that the sum is divisible by 2 in a few examples; it constructs the integer witness $r + s$.

4 Universal and existential statements

To prove $\forall x \in D\, P(x)$, take an arbitrary $x \in D$ and prove $P(x)$ without assuming anything special about x. To prove $\exists x \in D\, P(x)$, give a witness and verify it satisfies the predicate. To disprove a universal claim, one counterexample suffices. To disprove an existential claim, explain why no possible witness can work.

Worked example: counterexample versus proof

The statement "every odd integer is prime" is false: 9 is odd, but $9 = 3 \cdot 3$ is composite. This one counterexample disproves the universal statement. By contrast, showing that $3, 5, 7$ are prime does not prove the statement.

5 Closure properties and the right level of detail

Integers are closed under addition, subtraction, and multiplication, but not division. Rational numbers are closed under addition and multiplication, and division by a nonzero rational number. Whenever a proof constructs a new integer expression such as $3r - 2s$, cite integer closure before claiming that expression is an integer.

Worked example: why division needs care

Let a/b and c/d be rational, with $b, d \neq 0$. Their sum is $(ad + bc)/(bd)$. Integer closure shows $ad + bc, bd \in \mathbb{Z}$. Moreover $bd \neq 0$ because both b and d are nonzero. Therefore the sum is rational. Omitting the last sentence leaves an essential condition unverified.

6 Proof hygiene and common errors

Use $\in$ for membership and $\subseteq$ for subset inclusion. Do not confuse $=$ with $\equiv$ (logical equivalence) or $\equiv_m$ (congruence modulo m). Cite definitions at the moment they are applied. An implication proof should conclude with the desired implication, not simply with an isolated equation. A counterexample must satisfy the original hypothesis and violate the conclusion.

7 Practice problems

1. Prove that the sum of three even integers is even.
2. Disprove: every integer whose square is even is odd.
3. Prove that the product of an even integer and any integer is even.
4. Explain why a/b is not necessarily a rational number when $a, b \in \mathbb{Z}$ alone.
5. Rewrite "every integer has a greater integer" with quantifiers and prove it.
6. Disprove: if $4 \mid n$, then $8 \mid n$.

8 Solutions to practice

1. Write $a = 2r, b = 2s, c = 2t$. Then $a + b + c = 2(r + s + t)$ and $r + s + t \in \mathbb{Z}$ by closure.
2. $n = 2$ has even square 4, but 2 is not odd.

3. If $a = 2k$ and $b \in \mathbb{Z}$, then $ab = 2(kb)$; $kb \in \mathbb{Z}$ by closure.
4. The denominator may equal zero; $1/0$ is undefined. Require $b \neq 0$.
5. $\forall n \in \mathbb{Z}\, \exists m \in \mathbb{Z}\, (m > n)$. For arbitrary n, choose $m = n + 1$.
6. $n = 4$ satisfies $4 \mid 4$ but $8 \nmid 4$.